

5 Source Channel Separation and Feedback Channels

Reading: Chapter 7.12-13

Preview of Lec. 20

- Source-Channel Separation Theorem
- Feedback Channels

In the next lecture, we will cover a few important topics, each of which has some significant impacts in designing information transmission system, both from a practical point of view and conceptual level.

5.1 Source-Channel Separation Theorem

In this course, we considered source coding problem and channel coding problem, separately. Before midterm, we studied [source coding problem], where the goal is to compress a random source (a sequence of independent random variables V_i with distribution P_V) with minimum number of bits per source symbol, i.e., $H(P_V)$ bits/symbol. After midterm, we studied [channel coding problem], where the goal is to reliably transmit as many bits as possible over a channel W . The maximum number of bits that can be reliably transmitted over a channel W is equal to $C = \max_{P_X} I(P_X, W)$.

The reason we could consider source coding problem and channel coding problem separately is due to Shannon's source-channel separation theorem. This theorem is indeed the one of greatest Shannon's contributions to digital communication systems, and this theorem says that "when the goal is to reliably transmit a random source over a given channel, the best performance one can achieve can be achieved with separated source coding and channel coding." We will prove this below.

Consider the task of transmitting a discrete memoryless source over an arbitrary channel. Let the source be a length- n sequence V^n with i.i.d. symbols following the distribution P_V . Suppose we wish to reconstruct the source as a length- n sequence $\hat{V}^n$. We use a DMC(W) with input alphabet $\mathcal{X}$ and output alphabet $\mathcal{Y}$ total n times to transmit the source sequence V^n with the aim of reconstructing it as $\hat{V}^n$.

There is an obvious solution to this problem. We can first do a lossless source coding on DMS(P_V), reduce n source symbols into nR bits; and then do a channel coding to reliably transmit those nR bits to the receiver, which recovers those bits and then decodes and reconstructs the source. We can do this if the channel has a capacity C that is larger than $H(P_V)$, the minimum rate required to do lossless compression for the source $V \sim P_V$.

The following theorem states that such a separation of source coding and channel coding is actually optimal. This has tremendous impacts as now any source and any channel can be connected through the universal interface of "bit streams," without loss of optimality. The entire industry of digital communications is built upon this result. The precise statement of the theorem is given as follows.

Theorem 1 (Source-Channel Separation Theorem). *Given a DMS(P_V), suppose that the sequence V^n is encoded into an input sequence X^n to a DMC(W) with capacity C , for which the output Y^n is mapped to $\hat{V}^n$. The reliable transmission of the source*

$$P_e^{(n)} := \Pr(\hat{V}^n \neq V^n) \rightarrow 0 \text{ as } n \rightarrow \infty \quad (1)$$

is achievable if and only if

$$C > H(P_V). \quad (2)$$

Proof. We already proved the achievability by using the separated source coding and channel coding. What is intriguing about this theorem is actually the **converse part**, i.e., the “only if” part, since in general, one can directly map the source sequence V^n to the channel inputs X^n , instead of first mapping it into bits and then to channel inputs. The proof is no hard at all.

$$\begin{aligned}
nH(P_V) &= H(V^n) \quad \text{Since } V^k \text{ are i.i.d. with distribution } P_V \\
&= H(V^n | \hat{V}^n) + I(V^n; \hat{V}^n) \\
&\leq \log 2 + P_e^{(n)} \log |\mathcal{V}|^n + I(V^n; \hat{V}^n) \quad \text{By Fano's inequality} \\
&\leq \log 2 + P_e^{(n)} \log |\mathcal{V}|^n + I(X^n; Y^n) \quad \text{By Data Processing inequality} \\
&\leq \log 2 + P_e^{(n)} \log |\mathcal{V}|^n + nC.
\end{aligned} \tag{3}$$

By rearranging terms, we have

$$P_e^{(n)} \geq \frac{nH(P_V) - nC}{n \log |\mathcal{V}|} - \frac{\log 2}{n \log |\mathcal{V}|}. \tag{4}$$

The last term is negligible as $n \rightarrow \infty$. Note that if $H(P_V) > C$, then $P_e^{(n)}$ is bounded below by a positive number so that it is impossible to make $P_e^{(n)} \rightarrow 0$. Therefore, to make $P_e^{(n)} \rightarrow 0$, it is necessary that $C \geq nH(P_V)$. □

5.2 Feedback Channels

I also want to add a small section about feedback channels. Suppose that in a given DMC, we also have some signals sent back from the receiver to the transmitter. We assume here that **perfect causal feedback** is available. That is, at time i , the transmitter also observes the output signals Y_1^{i-1} . Our question is: can this feedback be of any help to the forward communications?

The encoding of such channel is a bit different. To take the advantage of the feedback information, the encoder is a time-varying map. At time i , we need an encoding function

$$f_i : \{1, 2, \dots, 2^{nR}\} \times \mathcal{Y}^{i-1} \rightarrow \mathcal{X} \tag{5}$$

to determine the next transmitted symbol $X_i(M, Y^{i-1})$. Clearly, the feedback capacity, which is defined as the maximum achievable data rate with feedback, is larger than or equal to the capacity without feedback

$$C_{\text{FB}} \geq C. \tag{6}$$

What is surprising is that if the underlying channel is DMC, the **feedback does not help increase the capacity**.

Theorem 2.

$$C_{\text{FB}} = C. \tag{7}$$

Proof. What we need is a more careful converse. Consider

$$\begin{aligned}
nR &= H(M) \\
&= H(M | Y^n) + I(M; Y^n) \\
&\leq \log 2 + P_e^{(n)} nR + I(M; Y^n) \quad \text{Fano's inequality.}
\end{aligned} \tag{8}$$

Here for the case without feedback we used to have

$$\begin{aligned}
I(M; Y^n) &\leq I(X^n; Y^n) \quad \text{Data processing} \\
&= H(Y^n) - \sum_i H(Y_i | X^n, Y^{i-1}) \\
&= H(Y^n) - \sum_i H(Y_i | X_i).
\end{aligned} \tag{9}$$

But now the first inequality is kind of loose and the last equality is no longer valid with feedback, since Y_i might be dependent on future inputs. To fix this, we write

$$\begin{aligned}
I(M; Y^n) &= H(Y^n) - H(Y^n|M) \\
&= H(Y^n) - \sum_i H(Y_i|M, Y^{i-1}) \\
&= H(Y^n) - \sum_i H(Y_i|M, Y^{i-1}, X_i) \quad \text{the key here is we do not give the entire } X^n \text{ but only } X_i \\
&= H(Y^n) - \sum_i H(Y_i|X_i) \\
&\leq \sum_i (H(Y_i) - H(Y_i|X_i)) \\
&\leq nC.
\end{aligned} \tag{10}$$

□

Remarks: Even though feedback does not help increase the capacity, it can dramatically reduce the difficulty in coding. Consider the BEC case. Moreover, with feedback much higher error exponent is achievable and hence the feedback can reduce the length of channel codes to achieve a certain decoding error threshold.